

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (ME) Examination

November 2013

CL 202 Strength of Materials

Date: 23.11.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the followings:

1. What is section modulus if a rectangular section having width (b) and depth (d)? [01]
2. What is section modulus of a hollow circular section if, d is inner diameter and D is outer diameter of circle? [01]
3. A beam of symmetrical section is 350 mm deep has a moment of inertia of $13160 \times 10^6 \text{ mm}^4$ about its principal axis. To what radius may it bent if the maximum bending stress is not to exceed 126 N/mm^2 . Consider $E = 2 \times 10^5 \text{ N/mm}^2$. Also find out the moment of resistance of the section at this stress. [02]
4. A beam of symmetrical section, depth = 400 mm, $I = 19300 \text{ cm}^4$ is simply supported over a span of 8 meters. What uniformly distributed load may it carry if the maximum bending stress is not to exceed 120 N/mm^2 ? What concentrated load may be carried by the beam at the centre with the same permissible bending stress? [03]

- Q - 2 (a)** A rectangular hole is made in a triangular section as shown in **Figure 1**. Determine the moment of inertia of the section about base BC. [04]
- (b)** **Figure 2** shows the cross-section of a cast iron beam. Determine the moment of inertia of the section about vertical axes (I_{yy}) passing through the centroid of the section. [05]
- (c)** A flitched beam consists of two wooden joist 150 mm wide and 300 mm deep and a steel plate 10 mm thick and 250 mm deep placed symmetrically between wood sections. If the maximum stress in the wooden joist is 10 N/mm^2 . Find the corresponding maximum stress attained in steel. Find also the moment of resistance of the section. Take $E_s = 20 E_w$. [05]

OR

Q - 2 (a) Determine the moment of inertia of the shaded area of a semi-circle as shown in **Figure 3** about its base AB. [04]

(b) Compute the moment of inertia of the cross-section as shown in **Figure 4** about axis AA. [05]

(c) Explain the term "Strength of a section". [05]

Determine the dimensions of joist of a timber for span 8 m to carry a brick wall 200 mm thick and 5 m high, if the density of brick work is 1850 kg/m^3 and the maximum permissible stress is limited to 7.5 MN/m^2 . Given that the depth of joist is twice the width.

Q - 3 Attempt any Two: [14]

(a) A small block 40 mm long $\times$ 30 mm high and 5 mm thick is subjected to uniformly distributed tensile forces having the resultant values shown in **Figure 5**. Compute the stress component developed along the diagonal AC.

(b) At a certain point in a strained material, normal stresses of 128 MN/m^2 , tension, and 100 MN/m^2 , compression are acting on planes right angles to each other. The greater principal stress in the material is limited to 160 MN/m^2 . To what shearing stress may the material be subjected on the given planes, and what will be the maximum shearing stress at that point?

(c) At a certain point in a strained material, the values of normal stresses across two planes at right angles to each other are 80 MPa and 32 MPa, both tensile and there is a shear stress of 32 MPa clockwise on the plane carrying 80 MPa stress across the planes as shown in **Figure 6**.

(i) Locate the principal planes and evaluate the principal stresses

(ii) Calculate the maximum shear stress and specify its plane

SECTION – II

Q - 4 Answer the followings:

1 In a triangular section, the maximum shear stress occurs at _____. [01]

(a) Apex of the triangle (b) mid of the height (c) $1/3$ of the height

2 When a rectangular section of a beam is subjected to a shearing force, the ratio of maximum shear stress to the average shear stress is _____. [01]

3 Draw qualitative shear stress distribution diagram of symmetrical I- Section and Diamond section. [02]

4 A solid shaft transmits a torque of 50 kNm. If the angle of twist is limited to 1.5° in 2 [03]

m length of shaft and the shear stress to 80 N/mm^2 , determine the diameter of shaft.
Take $G = 0.8 \times 10^5 \text{ N/mm}^2$

- Q - 5 (a) A close-coiled helical spring is required to carry a load of 200 N. If the mean coil diameter is to be 8 times that of the wire, calculate the diameter of wire and coil. Take maximum shear stress as 120 MPa. [03]
- (b) A composite shaft is made up of brass rod of 40 mm diameter enclosed in a steel tube of external diameter 50 mm and 5 mm thick. The shafts are rigidly fixed to each other so that there is no relative motion between the two. The shaft is required to transmit a torque of 4 kNm. Determine the stresses developed in brass and steel if both shafts have equal lengths. Take $G_{\text{brass}} = 40 \text{ GPa}$ and $G_{\text{steel}} = 80 \text{ GPa}$. [05]
- (c) A beam in the form of cross is shown in **Figure 7** is subjected to shear force of 25 kN. Draw the shear stress distribution diagram. [06]

OR

- Q - 5 (a) A solid circular shaft of 3 m length and 200 mm diameter is subjected to a torque of 100 kNm, find the relative rotation between the end cross-sections of the shaft. Given $C = 10^5 \text{ N/mm}^2$. [03]
- (b) A 60 mm diameter shaft transmits 80 kW at 100 rpm. The shaft is connected to a machine component by means of key which is 20 mm wide and 100 mm long. Find the shear stresses developed in the shafts and the key. [05]
- (c) A beam in the form of cross is shown in **Figure 8** is subjected to shear force of 100 kN. Sketch the shear stress distribution diagram. [06]
- Q - 6 Attempt any Two: [14]
- (a) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 9**.
- (b) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 10**.
- (c) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 11**.

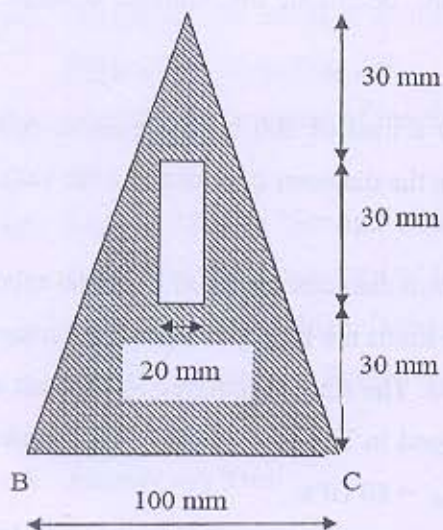


Figure 1

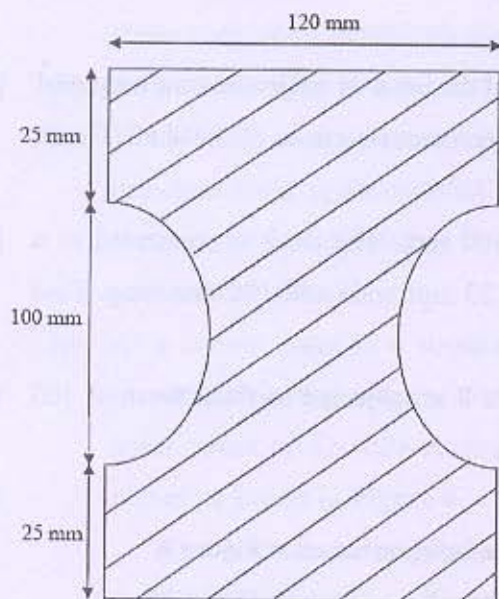


Figure 2

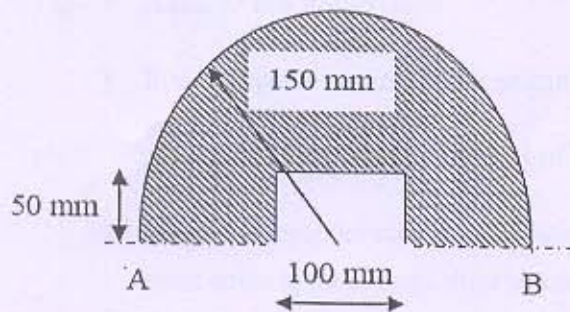


Figure 3

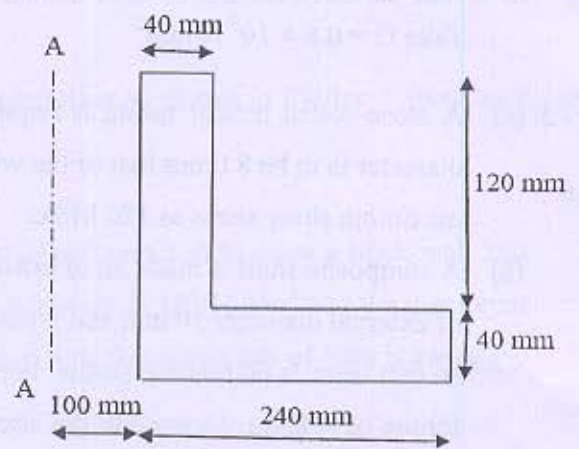


Figure 4

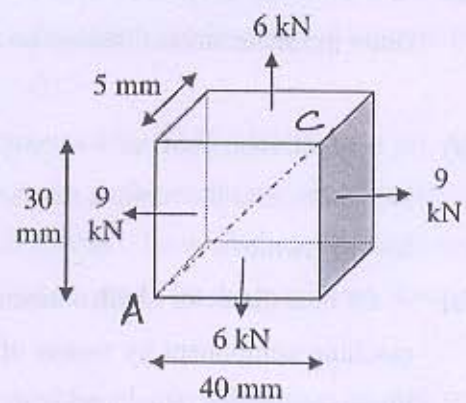


Figure 5

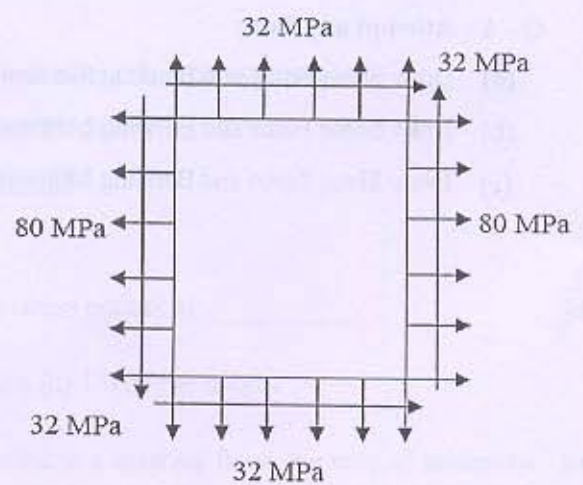


Figure 6

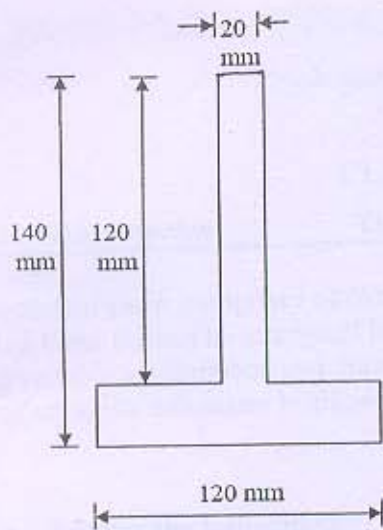


Figure 7

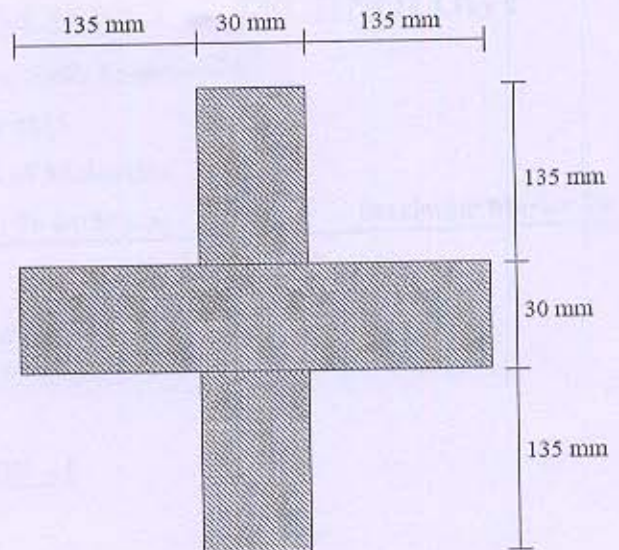


Figure 8

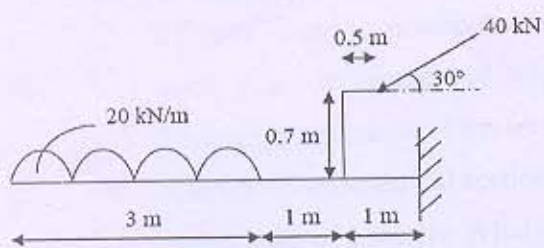


Figure 9

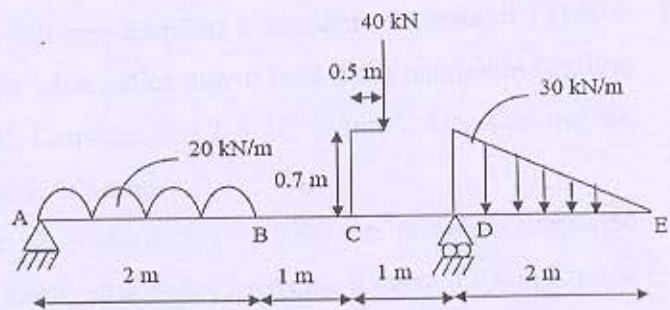


Figure 10

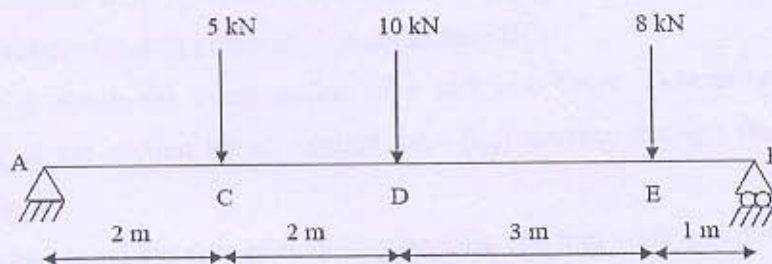


Figure 11